

# Saif Ur Rehman

*Machine learning engineer and AI researcher*

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## EDUCATION

**MSc Data Science and Artificial Intelligence**, University of Liverpool 2025 – 2026

Dissertation: Beyond AlphaFold: Filling the Blind Spots in Protein Structure Prediction, supervised by Dr Olga Anosova.

**BE Electrical (Telecommunication) Engineering**, National University of Sciences and Technology (NUST), Pakistan 2016 – 2021

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## RESEARCH

**Evidence and uncertainty in geometry foundation models** 2026 – present

Research in progress · Target: CVPR 2027 · First-author manuscript in development

When a feed-forward 3D/4D reconstruction model gives a confident answer, is that answer supported by the geometry in its input or by a learned prior that merely looks plausible? And could the model's uncertainty tell the two apart?

**Peer-reward misalignment in multi-agent LLMs** 2026

Under review · Submitted to a NeurIPS 2026 workshop

If two language-model agents grade each other and each is trained on the grade its peer assigns, is giving the solving and grading roles to different models enough to keep the training signal honest?

I wrote the single-agent RL training loop (REINFORCE with a moving-average baseline) that the two-agent trainer builds on, and produced the single-agent replication figure. In follow-up work I audited how policy-gradient credit reaches the grading tokens and designed the missing controls as a reward-topology comparison (objective reward, a frozen judge, a four-agent cycle, reciprocal pairs); it has not been run yet.

**Beyond AlphaFold: Filling the Blind Spots in Protein Structure Prediction** 2026

MSc dissertation · OpenFold retraining in progress

github.com/saifrhman/702\_BeyondAF

What changes when redundancy in the Protein Data Bank, the training data for AlphaFold2 and OpenFold, is measured exactly on backbone geometry across the whole archive, rather than by the usual sequence identity?

Sole author of the dissertation, supervised by Dr Olga Anosova (University of Liverpool). The invariant, the cleaning protocol and the multi-stage search strategy come from earlier BRI work on the PDB, and the OpenFold scripts started from a predecessor project. I re-implemented and scaled that protocol, implemented the compressed cover tree of Elkin and Kurlin (2023), and added the direct-edge removal policy, the provenance and validation layers, the sequence-redundancy comparison and the projection of retained chains onto OpenFold's training inputs.

**Geographical Data and Computational Tools for Determining Restoration Priorities in Mangrove Ecosystems: A Scoping Review** 2025 – 2026

Manuscript in preparation · Scoping review; began as MSc group coursework

Which remote-sensing data and machine-learning methods are used to decide where mangrove restoration is needed, how are they evaluated, and do they hold up outside the region they were developed in?

One of five co-authors, listed alphabetically by surname. I restructured the manuscript and drafted large parts of the sections on data sources, sensing modalities, computational methods and future work.

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## EXPERIENCE

**Applied AI Intern**, Rofsix 2026 – present

Developing a real-time system that detects and redacts confidential information during screen sharing, combining OCR, transformer-based NER and image processing. Built and evaluated inference components with ONNX, OpenVINO and PaddleOCR, and profiled the pipeline for low-latency CPU execution.

**Senior Systems Engineer**, RoshTech Nov 2022 – Sep 2025

Built threat-detection and anomaly-analysis workflows over MongoDB, Graylog and the Elastic Stack to support incident investigation for enterprise clients.

**Data Science Intern (part-time), Creative Tech Solutions**

Jan 2024 – Apr 2024

Wrote Python scripts for data analysis and data cleaning for the analytics team.

**Senior Cloud Engineer, Tier3tech**

Feb 2022 – Nov 2022

Worked on cloud security and Azure Virtual Desktop infrastructure; mentored team members.

**Level 2 Support Team Lead, MiGo Innovations**

Nov 2020 – Feb 2022

Led a team running offshore network operations for US clients.

**SELECTED PROJECTS**

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**F1 AI Copilot**[github.com/saifrhman/f1-ai-copilot](https://github.com/saifrhman/f1-ai-copilot)

Question answering over the 2026 FIA F1 regulations (592 pages) that checks each answer's citations, rule identifiers and numbers against the cited passages (inference-marked sentences are exempt from the number check) and declines when they do not match.

**Hybrid receipt extraction**[github.com/saifrhman/noise-robust-ocr-pipeline](https://github.com/saifrhman/noise-robust-ocr-pipeline)

Receipt field extraction combining EasyOCR, a rule parser and LayoutLMv3 trained on source-weighted weak labels; no evaluation results committed yet.

**Football intelligence platform**[github.com/saifrhman/football-analytics-platform](https://github.com/saifrhman/football-analytics-platform)

Bronze/silver/gold pipeline from StatsBomb Open Data to BigQuery with dbt models, 23 schema tests and a FastAPI/Streamlit serving layer; run end to end on a five-match sample.

**TacticLens**[github.com/saifrhman/tacticlens](https://github.com/saifrhman/tacticlens)

Research design and data foundations for structural retrieval over football tracking data; retrieval model not yet built.

**Closed-loop Bayesian optimisation**[github.com/saifrhman/closed-loop-hpo-botorch](https://github.com/saifrhman/closed-loop-hpo-botorch)

Gaussian-process Bayesian optimisation in BoTorch compared with random search for random-forest tuning (33 evaluations, single seed).

**COMPETITIONS**

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**ICPC Northwestern Europe Regional Contest (NWERC) 2025:** 107th, as the University of Liverpool team “Baby Harp Seal”.

**ICPC UK & Ireland Programming Contest 2025:** 50th overall and 1st at the University of Liverpool site (team event).

**Liverpool Computer Society Hackathon:** 1st place (team).

**Ultamation Hackathon:** 2nd place, constraint-satisfaction scheduling.

**SKILLS**

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**Research and ML:** PyTorch · RL fine-tuning of LLMs (LoRA, REINFORCE) · Retrieval-augmented generation · Weak supervision · Bayesian optimisation (BoTorch, GPyTorch) · Experimental design and error analysis

**Geometry, vision and science:** 3D/4D reconstruction models · Protein structure data (OpenFold, MMseqs2, gemmi, Mol\*) · Exact metric search (cover trees, k-d trees) · Document AI (LayoutLMv3, EasyOCR) · OpenCV

**Systems and data:** Python, C++, SQL · FastAPI, Streamlit · Qdrant, LangChain · BigQuery, dbt · Slurm and HPC clusters · Docker, GitHub Actions, Linux · ONNX, OpenVINO